



College of Business

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FinTech and AI in Finance

Class 4: Nonlinear Machine Learning for Return Prediction

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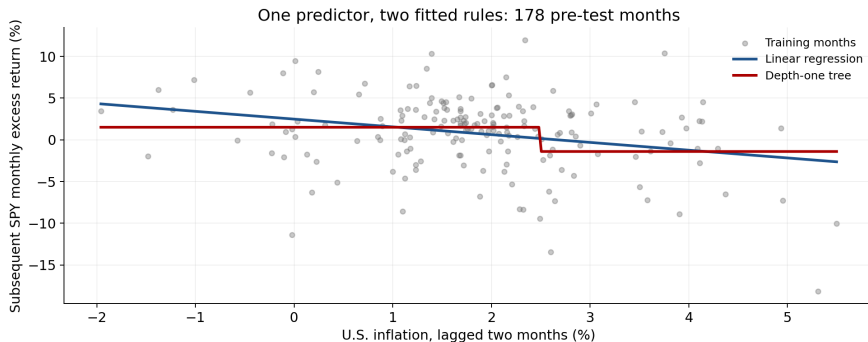
Class 3 used linear combinations of predictors. Here we allow a signal's effect to change with its level or with other characteristics.

- ▶ **Trees:** forecasts differ across observable regimes.
- ▶ **Forests and boosting:** combine trees to stabilize or refine a forecast.
- ▶ **Neural networks:** learn combinations of signals and their interactions.

We compare market excess-return forecasts and weekly stock excess-return forecasts using Class 3's dates and benchmarks.

Regimes and Nonlinear Forecasts

Linear and Tree Forecasts from the Same Data



Keep in mind

A line uses one slope; a tree uses discrete regimes. The fitted rule must predict well on later data.

Regression Trees: Forecasts by Regime

A tree divides observations into groups called **leaves**. In leaf R_ℓ , the forecast is the mean training return:

$$\hat{c}_\ell = \frac{1}{n_\ell} \sum_{i \in R_\ell} y_i.$$

n_ℓ : number of training observations in the leaf.

- ▶ **When useful:** the relation changes at a threshold or depends on another predictor.
- ▶ **How to forecast:** follow the fitted splits into one leaf and use its mean return.
- ▶ **Complexity:** depth and minimum leaf size limit how finely the tree divides the data.

Regression Trees: Fitting and Using the Rule

At each node, choose predictor j and threshold s to minimize:

$$(j^*, s^*) = \arg \min_{j, s} \left[\sum_{i \in R_L(j, s)} (y_i - \bar{y}_L)^2 + \sum_{i \in R_R(j, s)} (y_i - \bar{y}_R)^2 \right].$$

R_L : node observations with $x_{ij} \leq s$; R_R : those with $x_{ij} > s$. Each $\bar{y}$ is the corresponding training mean.

1. Repeat within each child node, subject to depth and leaf-size limits.
2. Choose those limits on chronological validation data; refit before testing.
3. Route a new observation using its predictors only.

Cross-Sectional Splits and Time-Series Splits

Cross-section: split stocks

In the same month, separate firms by a characteristic such as size.

Different stock groups can have different return relations.

Time series: split months

For the same market, separate months by a state such as lagged inflation.

Different market states can have different return relations.

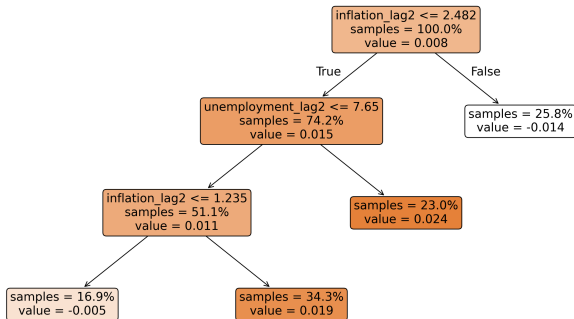
Asset Heterogeneity and Uncommon Factors

The paper combines stock-group trees with variable selection: a characteristic can enter the return model for some groups but not others. Separate expansion and recession fits compare these patterns across business-cycle states.

Cong, Feng, He, Li, and Zhang (2026 working paper). The business-cycle comparison uses retrospective NBER dates.

The U.S. Macro Tree Uses One Inflation Split

Validation-tuned U.S. macro decision tree



The validation-tuned tree separates low- and high-inflation months. Test data assess its forecasting value.

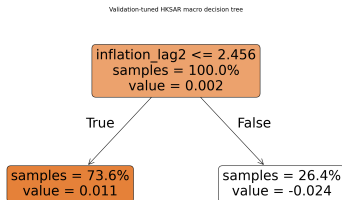
How to Read the First Split

The validation-tuned U.S. tree asks whether two-month-lagged inflation is at most 2.48%.

Training region	Observations	Leaf forecast
Inflation $\leq 2.48\%$	132 months	+1.5%
Inflation $> 2.48\%$	46 months	-1.4%

The threshold is a fitted forecast rule, not an estimate of the causal effect of inflation.

Similar Inflation Thresholds, Different Market Outcomes



Validation again picks depth 1 with 36-month leaves, and the root again splits on U.S. inflation at 2.46%. The 131 training months at or below it forecast +1.1% for the Hang Seng; the 47 months above it forecast -2.4%.

Keep in mind

The U.S. tree has test OOS $R^2 = -2.6\%$ versus MA(12), compared with +4.7% for HKSAR. Similar fitted thresholds can have different forecasting value.

Regression Trees: Choosing Depth on Validation Data

Suppose we fit trees with different depth limits on the same development months:

Maximum depth	1	2	4	8
Training MSE $\times 10^4$	24	21	16	9
Validation MSE $\times 10^4$	28	26	29	35

Choose depth two. Deeper trees fit the development returns more closely but make worse forecasts on the validation months.

Refit at the chosen depth before the test. Minimum leaf size provides a second way to limit how finely the tree divides the sample.

Illustrative errors, not results from the course dataset.

Tree Depth and Financial Regimes

- ▶ **Depth one:** one question gives two return forecasts.
- ▶ **Depth two:** a second question can distinguish momentum behavior within a volatility regime.
- ▶ **Greater depth:** narrower groups describe more local patterns, using fewer observations per group.

Validation chooses how many distinctions the data can support.

Leaf Size Sets the Evidence Required for a Regime

- ▶ **Small leaf:** can represent a rare crisis or liquidity state, but its average return is noisy.
- ▶ **Large leaf:** makes each regime estimate more stable, but may hide a real rare state.
- ▶ **In our market exercise:** candidate minimum leaf sizes are 12, 24, and 36 months.
Validation chooses among them.

Keep in mind

A high-return leaf based on a few months is a noisier estimate than a similar leaf observed repeatedly.

Worked Example: Predicting with Two Leaves

Six training months are split at lagged inflation of 2.5%:

Inflation	1.0%	1.8%	2.2%	3.0%	3.6%	4.4%
Next-month return	3%	1%	2%	0%	-1%	-2%

$$\hat{r}_{\text{low}} = (3 + 1 + 2)/3 = 2\%, \quad \hat{r}_{\text{high}} = (0 - 1 - 2)/3 = -1\%.$$

A new month with lagged inflation of 2.0% receives a 2% forecast. At 3.5% inflation, the forecast is -1%.

The lagged inflation observation determines the branch. The leaf average comes from training returns.

Illustrative training observations.

Worked Example: Momentum Depends on Volatility

Suppose a fitted model gives these weekly excess-return forecasts:

Market state	Low momentum	High momentum	High minus low
Low volatility	0.0%	0.6%	+0.6%
High volatility	0.0%	−0.2%	−0.2%

The momentum effect changes sign with volatility. A regression with only additive momentum and volatility terms cannot reproduce this table.

A tree can first split on volatility, then on momentum within each branch.

Illustrative fitted forecasts, not estimates from the course panel.

Using an Interaction in Stock Selection

In the preceding example, a high-momentum stock receives:

$$\hat{r}^e = 0.6\% \text{ in low volatility,} \quad \hat{r}^e = -0.2\% \text{ in high volatility.}$$

A long-only manager might overweight the first stock and underweight the second.

- ▶ The same momentum screen now produces different positions across states.
- ▶ The interaction is useful if it improves later rankings or portfolio returns.

Class 5 considers how these forecasts become constrained weights.

Random Forests and Boosting

Bagging: Why Average Fitted Trees?

Fit B trees independently on bootstrap samples drawn from training data with replacement:

$$\hat{f}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}_b(x).$$

For tree predictions with variance σ_f^2 and common pairwise correlation ρ ,

$$\text{Var}\left(\frac{1}{B} \sum_b \hat{f}_b(x)\right) = \sigma_f^2 \left[\rho + \frac{1-\rho}{B} \right].$$

Why average: more trees reduce the uncorrelated part of prediction variance. Shared errors remain.

Resampling uses training data only; validation dates stay later in time.

Bagging: From Resampled Data to One Forecast

Draw several bootstrap training samples and fit one tree to each. Send the **same new observation** through every tree.

Suppose three trees forecast next month's return as

$$0.8\%, \quad -0.2\%, \quad 0.3\%.$$

Their bagged forecast is

$$\hat{r} = \frac{0.8 - 0.2 + 0.3}{3}\% = 0.3\%.$$

Choose tree settings on later validation dates. Averaging cannot remove shared forecast errors.

Illustrative forecasts.

Random Forests: Adding Random Predictor Selection

A random forest combines bootstrap sampling with random predictor subsets.

- ▶ Fit trees to different bootstrap samples of the training rows.
- ▶ At each node, search for the best split among a fresh random subset of predictors. Our model considers 5 of the 30 predictors.
- ▶ Average the B fitted trees' forecasts for a new observation:

$$\hat{f}_{\text{RF}}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}_b(x).$$

Random predictor selection can reduce correlation between trees, making averaging more effective.

Random Forests: Fitting a Weekly Stock Model

Start with the thirty lagged stock characteristics and next-week excess returns.

1. Draw bootstrap samples from the development rows. At each split, consider a random subset of characteristics.
2. Compare tree depth and minimum leaf size on the next 26 validation weeks, keeping all stocks from each week together.
3. Refit the chosen settings on the 104 pre-test weeks and average the trees' forecasts for each test stock.

Our implementation considers $\lfloor \sqrt{30} \rfloor = 5$ characteristics per split. It uses 300 trees during validation and 500 in the final refit.

Gradient Boosting: Successive Forecast Corrections

Fit corrections sequentially.

$$F_0 = \bar{y}_{\text{train}},$$

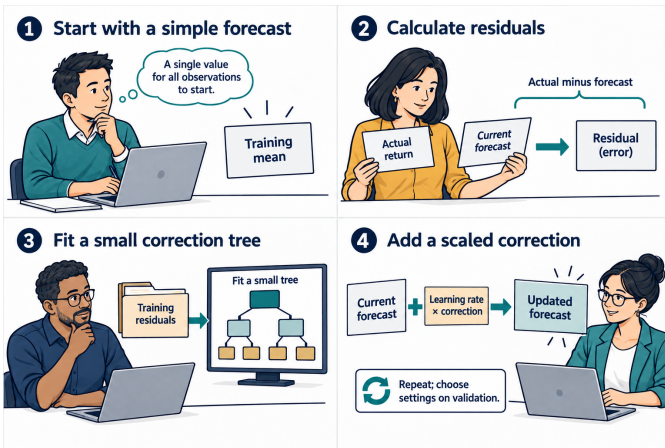
$$e_{im} = y_i - F_{m-1}(x_i),$$

$$F_m = F_{m-1} + \nu h_m.$$

h_m : small tree fitted to current training residuals.

ν : learning rate. Choose it, tree count, and depth on validation data.

A forest averages trees; boosting adds successive corrections.



Gradient Boosting: Calculating One Update

Start from the sample mean of the same six months, $F_0 = 0.5$, and take the residuals:

Return	+3	+1	+2	0	-1	-2
Residual $r_t - F_0$	+2.5	+0.5	+1.5	-0.5	-1.5	-2.5

Fit one stump to those residuals. The 2.5% threshold wins again, with leaf means $+1.5$ and -1.5 . With a learning rate $\nu = 0.1$:

$$F_1 = 0.5 + 0.1 \times (\pm 1.5) = 0.65 \text{ or } 0.35.$$

Later trees fit the remaining errors. Validation selects the number of updates.

Illustrative returns in percentage points, using the earlier six-month example.

Tune Learning Rate and Tree Count Together on Validation

- ▶ Pre-specify a small grid of learning rates and tree counts.
- ▶ Compare each pair on the same chronological validation sample.
- ▶ Prefer a stable region of good performance over one isolated minimum.
- ▶ Freeze the pair before opening the test period.

Early stopping is the same decision made one tree at a time: track validation loss as trees are added and stop when it no longer falls.

Random Forest or Gradient Boosting?

	Random forest	Gradient boosting
Construction	fits many randomized trees	fits corrections sequentially
Useful when	a single tree is unstable	a simple forecast leaves systematic errors
Forecast	average of the trees	sum of small corrections
Main control	depth, leaf size, feature sampling	learning rate, tree count, depth

The investment use is the same: forecast a return or rank stocks. Compare validation performance under the same target and information set.

Short-Answer Question: Is This Threshold Investable?

A tree splits at inflation = 2.5%. Pre-test diagnostics:

Diagnostic	Evidence
Leaf support	Only four months support the high-inflation leaf.
Leaf-size sensitivity	Minimum leaf size 12 removes the split.
Time stability	A later pre-test refit gives a different split.

1. Identify two warning signs in the diagnostics.
2. Explain whether the threshold is credible enough to guide a financial decision.
3. Propose one additional validation check without looking at the frozen test.

Interpretability alone does not establish a stable regime.

Multiple-Choice Question: Reading a Regression Tree

Suppose a fitted tree uses the following weekly forecasting rule:

Condition	Forecast return
Volatility $> 2\%$	-0.8%
Volatility $\leq 2\%$, momentum > 0	$+0.6\%$
Volatility $\leq 2\%$, momentum ≤ 0	-0.2%

A stock has volatility 1.5% and momentum -0.4% . What is its forecast?

- A) -0.8%
- B) -0.4%
- C) -0.2%
- D) $+0.6\%$

Multiple-Choice Question: Why Randomize Predictor Selection?

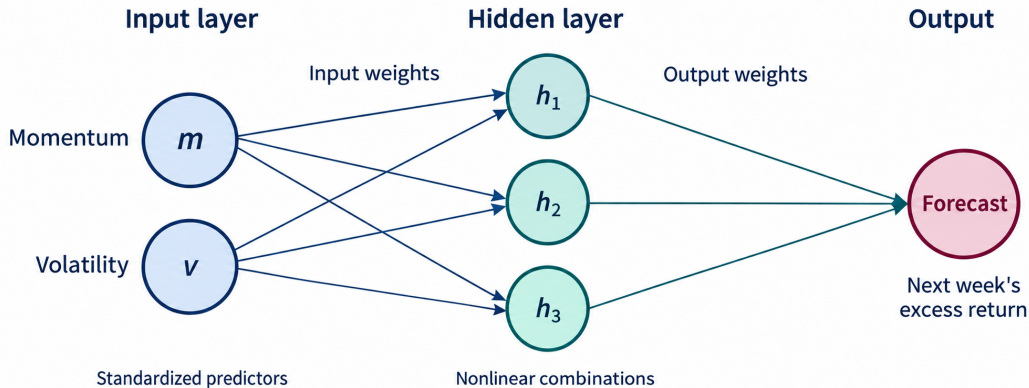
An analyst averages many trees fitted to bootstrap samples. Nearly every tree uses the same dominant predictor for its first split, and their forecast errors are strongly correlated.

Why might random predictor subsets improve the ensemble?

- A) They ensure that each individual tree has a lower training error.
- B) They can reduce correlation among tree forecasts, making averaging more effective.
- C) They ensure that the strongest predictor appears at the root of every tree.
- D) They remove the need to choose complexity on validation data.

Neural Networks and Market Evidence

Neural Networks: Learning Nonlinear Combinations



Hidden units apply a nonlinear activation to weighted inputs. Training learns the connection weights and intercepts.

Neural Networks: Fitting a Small Return Model

Estimate network parameters θ using training excess returns:

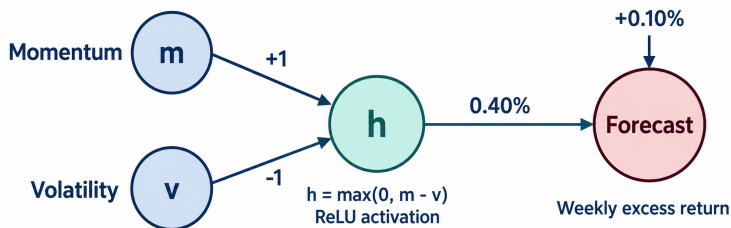
$$\hat{\theta} = \arg \min_{\theta} \left\{ \frac{1}{2n} \sum_{i=1}^n (a_i - f_{\theta}(z_i))^2 + \frac{\lambda}{2} \sum_{w \in \mathcal{W}} w^2 \right\}.$$

$\mathcal{W}$: connection weights; biases are unpenalized. This combines squared error with weight shrinkage.

1. Fit scaling and candidate networks on development weeks.
2. Choose hidden units, penalty, and training duration on later validation weeks.
3. Refit on pre-test weeks for the selected duration; evaluate once on the test.

Compare random initializations during development, before opening the test.

Worked Example: Reading a Hidden Unit



Stock	m	v	h	$\hat{r}^e$
A	1.0	0.5	0.5	0.30%
B	1.0	1.5	0.0	0.10%

Higher volatility lowers this forecast, holding momentum fixed.

Illustrative weights. Inputs m and v are standardized.

Early Stopping: Reading Training and Validation Loss

Suppose a neural network produces the following losses as training continues:

Training epoch	Training MSE	Validation MSE
10	0.0025	0.0027
30	0.0021	0.0023
60	0.0018	0.0026

The 30-epoch fit has the lowest validation error. Additional training lowers training error but worsens prediction on later observations.

Select the stopping rule before final testing. Use chronological validation with every stock from each date kept together.

Illustrative losses, not course-panel neural-network results.

Panel Size and the Number of Independent Dates

Course example	Usable observations	Distinct dates
Market prediction	238	238 months
Stock prediction	12,281	156 weeks

Stocks in the same week share news, funding conditions, and market shocks. Thousands of rows therefore provide fewer independent market episodes than the row count suggests.

Neural networks illustrate a possible extension here. Our reported course results use linear models and tree ensembles.

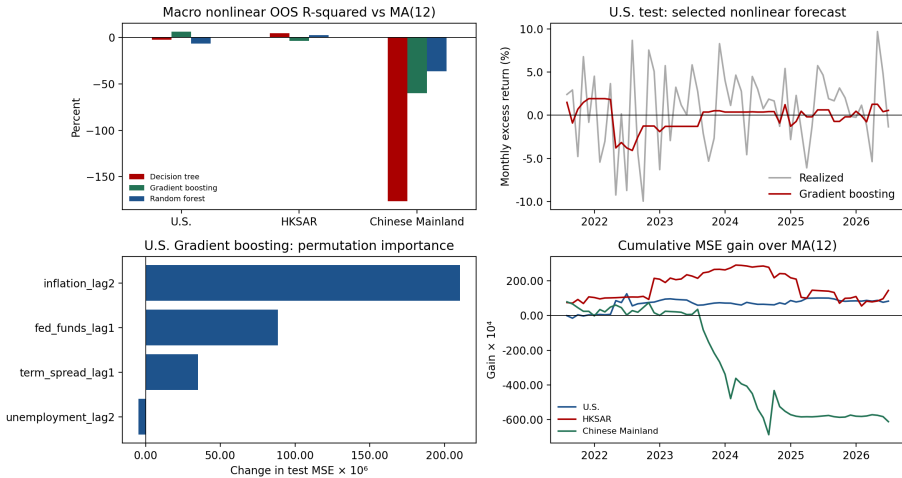
Short-Answer Question: How Should We Split a Stock Panel?

An analyst has 30 stocks observed over 156 weeks: 4,680 stock-week rows. She randomly splits the rows into training and test sets, then argues that the large sample justifies a large neural network.

1. Why do 4,680 rows not represent 4,680 independent market episodes?
2. Why is this random split unsuitable for evaluating forecasts of future returns?
3. Describe a better sample split and how you would choose the network size.

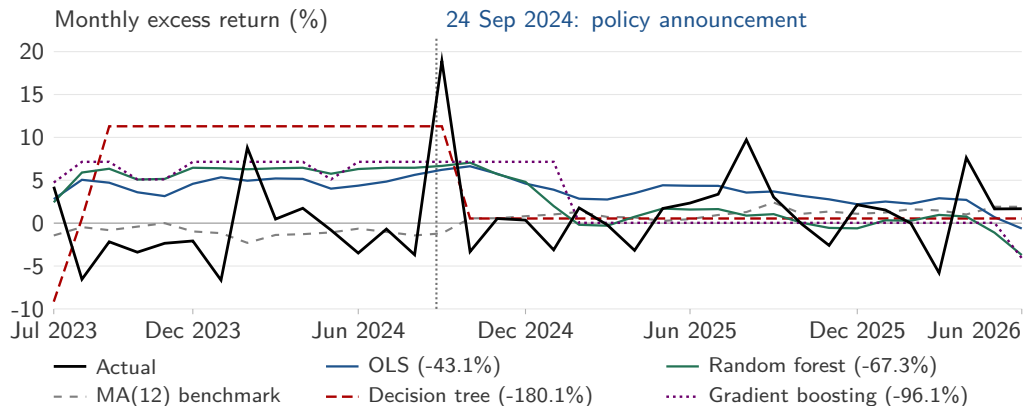
Nonlinear Market Excess-Return Forecasts Struggle Out of Sample

Nonlinear ML first tackles the same four-predictor market problem



CSI 300: Policy Changes and Return Prediction

July 2023–June 2026, 36 monthly excess-return forecasts.



Models retain their June 2021 fits. Legend values: OOS R^2 relative to MA(12).

Linear and Nonlinear Market Models on a Common Test

Market	Linear selection	OOS R^2	Nonlinear selection	OOS R^2
U.S.	LASSO	-16.3%	Gradient boosting	+6.3%
HKSAR	Elastic Net	+8.4%	Decision tree	+4.7%
Chinese mainland	LASSO	-0.6%	Random forest	-36.4%

Each entry uses the family selected on that class's validation sample. Full test: July 2021–June 2026, 60 months per market, against lagged MA(12).

Added flexibility does not improve all three markets. The comparison depends on the prediction problem and the available sample.

What the Market Results Mean for an Investor

The U.S. macro predictors describe economic conditions, but their relationship with subsequent equity returns changes across markets and periods.

- ▶ A readable inflation split is easy to explain, yet may forecast poorly.
- ▶ A flat, strongly regularized forecast can outperform a detailed regime rule.
- ▶ Class 5 tests whether these forecasts change market exposure profitably.

Nonlinear Stock Return Forecasts

Keep the Validation Logic; Change the Observation

Element	Macro-to-market	Characteristics-to-stock
Rows	238 months	12,480 grid rows; 12,281 usable rows
Features	four lagged macro variables	30 lagged technical ranks
Target	monthly market excess return	weekly stock excess return
OOS R^2 benchmark	lagged MA(12) excess return	zero excess return
Development	142 months	78 calendar weeks
Validation	36 months	26 calendar weeks
Test	60 months	52 calendar weeks

Keep in mind

Market models compete with a feasible MA(12) forecast; stock models compete with zero excess return. Every stock from the same week stays together.

How Much Data Support a Stock-Tree Leaf?

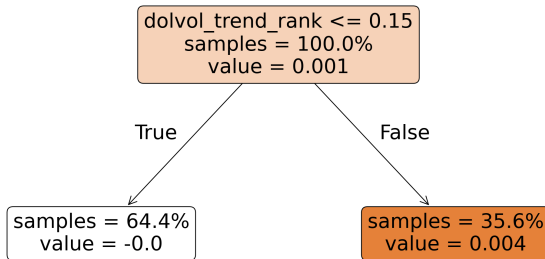
Candidate minimum leaf sizes: 30, 60, or 120 training stock-weeks.

- ▶ A leaf groups stock-weeks with similar characteristics. Its observations can be spread across many dates.
- ▶ Several stocks from one week can share the same return shock.
- ▶ Count the distinct weeks supporting a rule as well as the number of stock-week rows.

The validation exercise keeps all stocks from the same week together.

The Stock Tree Splits on Dollar-Volume Trend

Validation-tuned Hang Seng Index tree for excess returns



Follow the dollar-volume-trend rank to the two weekly forecasts. This is the validation-tuned tree candidate, not the overall winning nonlinear family.

Reading the Dollar-Volume-Trend Split

The fitted tree splits the centered dollar-volume-trend rank at about 0.150, roughly the 65th percentile.

Training group	Share of rows	Weekly excess-return forecast
Rank $\leq$ 0.150	64.4%	-0.01%
Rank $>$ 0.150	35.6%	+0.43%

The rule associates stronger trading-activity trends with higher subsequent returns in the fitted sample. Its test OOS R^2 is -0.1% against zero.

The displayed tree is the validation-tuned tree candidate. Random forest is the selected nonlinear family.

Validation Separates Three Nonlinear Families by Very Little

Family	Depth	Min. leaf	Rate	Trees	Validation MSE $\times 10^4$
Decision tree	1	30	–	1	26.30
Random forest	2	60	–	300	26.25
Gradient boosting	1	30	0.01	150	26.64

Keep in mind

Validation selects Random forest, but all three losses are close. Report the numbers and treat small differences cautiously; the table alone provides little basis for a strong claim that one nonlinear family is superior.

Course calculation using the fixed-vintage weekly stock panel.

Multiple-Choice Question: Which Nonlinear Model Was Selected?

Three models are tuned before the frozen test period. Their mean squared errors are:

Model	Validation MSE	Test MSE
Shallow tree	0.0024	0.0019
Random forest	0.0021	0.0023
Boosting	0.0022	0.0018

Which model should be reported as the **preselected evaluation model**?

- A. Shallow tree, because it has the lowest test MSE among the trees.
- B. Random forest, because it has the lowest validation MSE.
- C. Boosting, because it has the lowest test MSE.
- D. Average the tree and boosting after observing the test results.

Discuss

Separate model selection from final evaluation. Why cannot the test-period winner replace the model frozen after validation?

The Validation Winner Is Essentially Tied with Zero in the Test

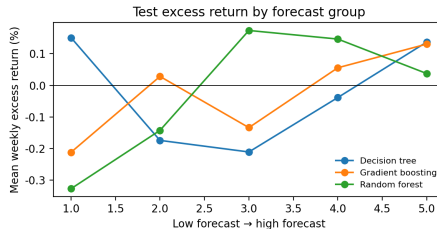
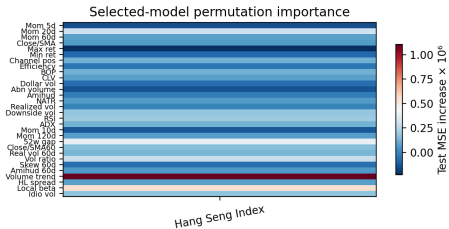
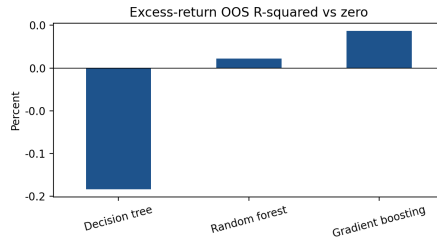
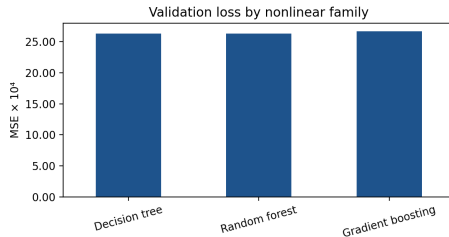
Frozen model	Test MSE $\times 10^4$	Test OOS R^2
Random forest (selected)	21.89	+0.0%
Gradient boosting	21.88	+0.0%
Decision tree	21.92	-0.1%

Do not switch after the test

Random forest remains the validation-selected model. Its test OOS R^2 is only slightly positive, so the economic evidence remains weak.

Nonlinear Stock Return Forecasts Also Remain Weak

Nonlinear ML on the Hang Seng Index stock panel: validation, test, importance, and sort



Worked Example: Read One Weekly Nonlinear Forecast

For Techtronic Industries (0669.HK) in the week ending 31 October 2025, returns rounded to one decimal place:

$$\underbrace{0.3\%}_{\hat{r}_{i,t}^{e, \text{RF}}} \quad \text{versus} \quad \underbrace{-8.4\% - 0.1\% = -8.5\%}_{r_{i,t}^e \text{ realized}}$$

Keep in mind

The selected model misses both sign and magnitude here. Evaluate it over all 52 test weeks, not this observation alone.

Course calculation using the fixed-vintage weekly stock panel.

Linear and Nonlinear Stock Return Forecasts Compared

Model	Role	Test MSE $\times 10^4$	OOS R^2
PCR	linear selection	21.89	+0.02%
Random forest	nonlinear selection	21.89	+0.01%
Gradient boosting	other candidate	21.88	+0.04%

Both classes use the same 52 test weeks and 4,108 stock-week outcomes. PCR is selected within the linear families, and Random forest within the nonlinear families.

In this panel, the extra flexibility does not deliver a larger test gain. It changes which characteristics drive the fitted forecast.

Stock Ranking Evidence for the Nonlinear Models

Model	Mean weekly top–bottom spread	<i>t</i> -statistic
Decision tree	-0.01%	-0.06
Random forest	+0.36%	1.04
Gradient boosting	+0.34%	0.87

Positive ensemble spreads have small *t*-statistics. Ranking and MSE evaluate different forecast properties.

52 test weeks. Equal-weight top-minus-bottom quintiles, unit exposure per side. Transaction costs ignored.

Accuracy, Ranking, and an Investment Mandate

Use of the forecast		Relevant evidence	What it assesses
Expected-return estimate	esti-	MSE, OOS R^2	numerical forecast accuracy
Stock ranking		group returns, top–bottom spread	ordering of subsequent returns
Long-only portfolio		excess return, risk, concentration	value after position constraints

A model can perform differently under these objectives. Choose the objective on development and validation data before the final test.

Short-Answer Question: Forecast Accuracy and Stock Ranking

Suppose a stock excess-return model has a test OOS R^2 of -1% against a zero forecast. Its top-minus-bottom portfolio has a mean weekly return of $+0.08\%$ before costs over the same test period.

1. Explain how negative OOS R^2 and a positive mean portfolio return can coexist.
2. Is the positive mean return enough to recommend investing? Give two additional checks and explain what each would tell you.

Interpreting the Weak Test Gains

Test gains from additional flexibility are limited.

- ▶ Thirty technical indicators overlap substantially in the information they contain.
- ▶ A fitted threshold may describe a relationship concentrated in a short period.
- ▶ More flexibility cannot compensate for a missing predictive variable.

The next research question concerns the information set and the stability of the relationship, as well as the algorithm.